

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 21, 2026

Where cancers are caught, and whether anyone looks

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

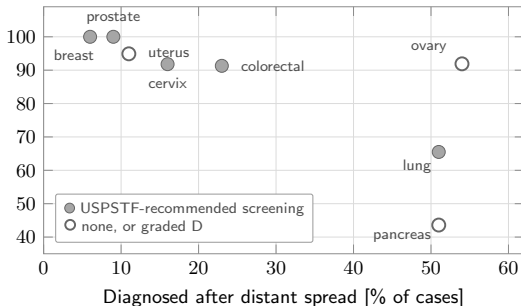
Vision

From currents to
a score

References

Backup

5-year survival if caught localized [%]



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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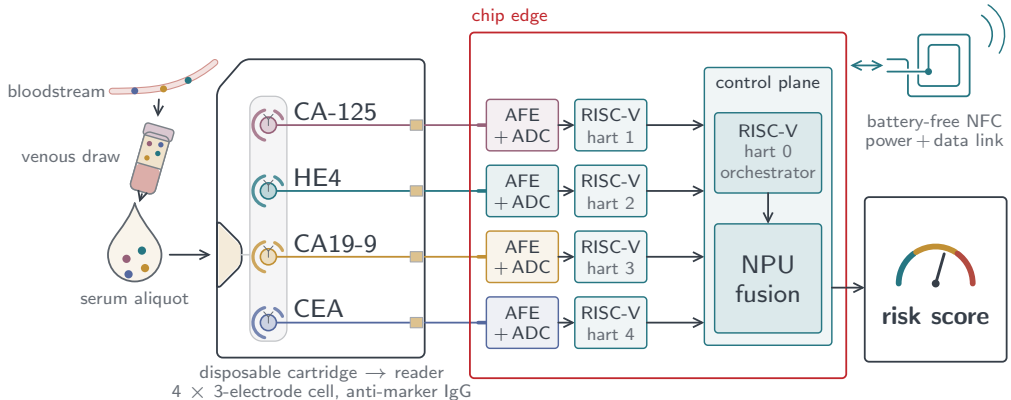
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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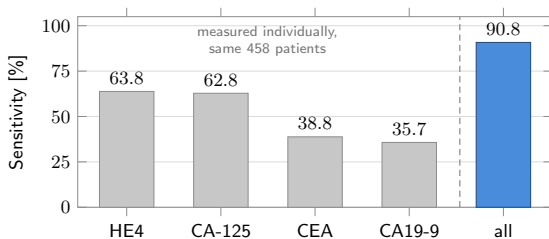
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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Block diagram

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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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ARTWORK PENDING



**ARTWORK
PENDING**



Grounded-WE two-amplifier potentiostat

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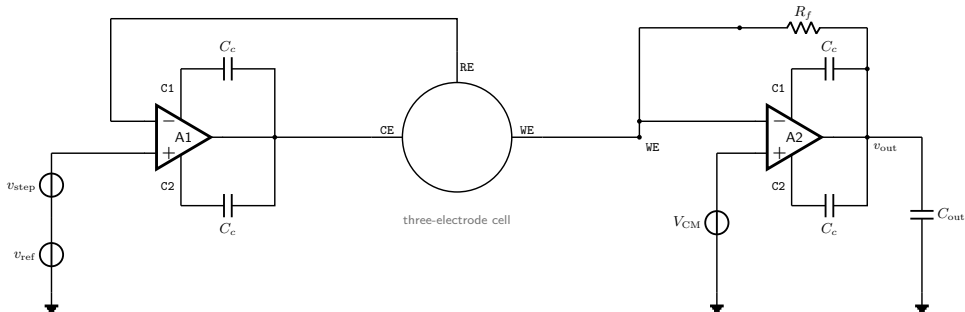
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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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Virtuoso export of the analog top level: four channels, shared bias, test bus

The electrode model, in Verilog-A

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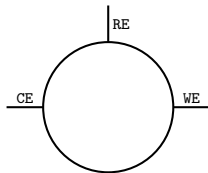
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R_u , R_{ce} , C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the four markers

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Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Channel range, assay range, and where the panel was placed

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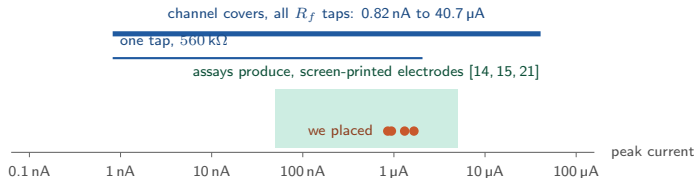
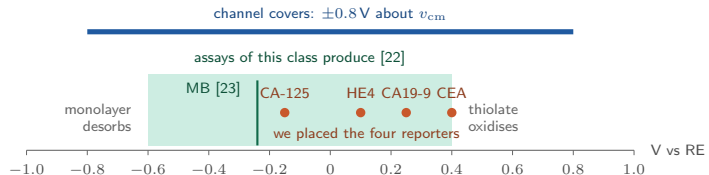
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Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Exercising the cell: differential-pulse voltammetry

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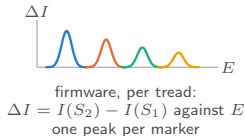
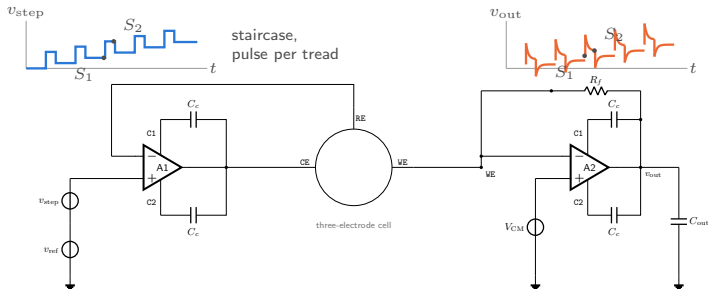
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Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

Resolve

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

Four markers resolved in one differential-pulse sweep

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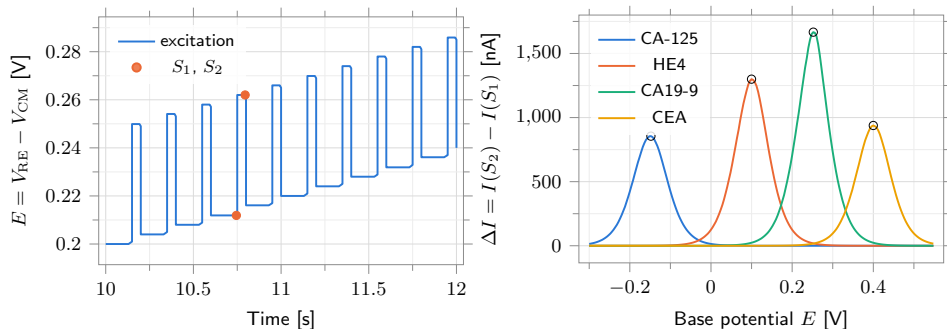
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Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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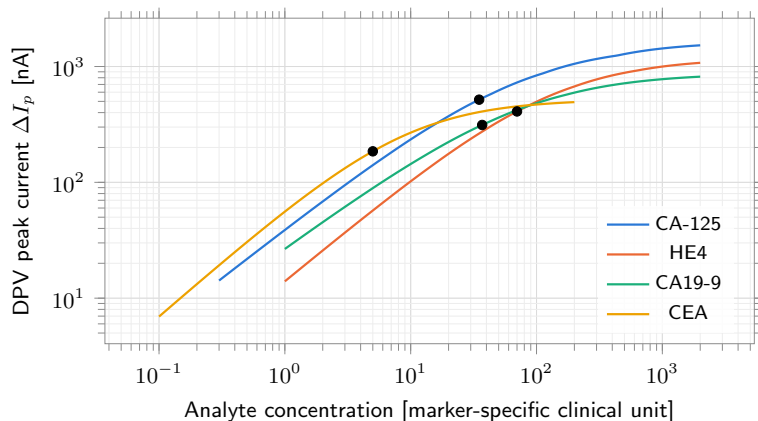
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CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic voltammetry through the assembled channel

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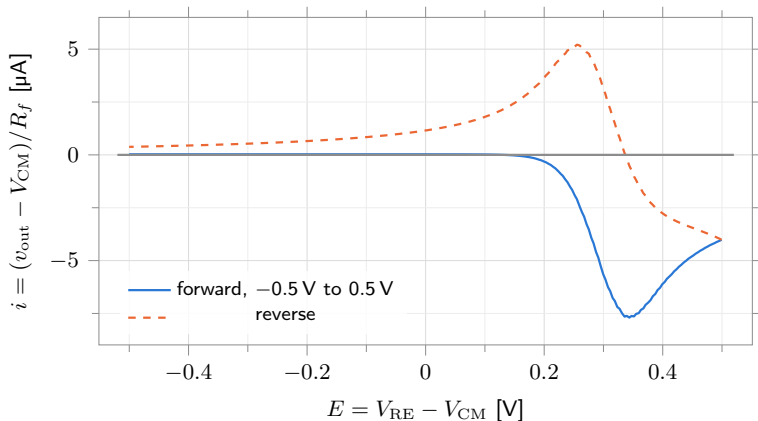
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Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Why impedance: for electrodes that stay in

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cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives
Firmware

each site's $R_{ct} \parallel C_{dl}$ arc, trended over the electrode's life, from the same channel
re-ranges, recalibrates, or retires the site

Drift is systematic: averaging removes noise, not bias [25].

How impedance is measured: the same loop, no extra hardware

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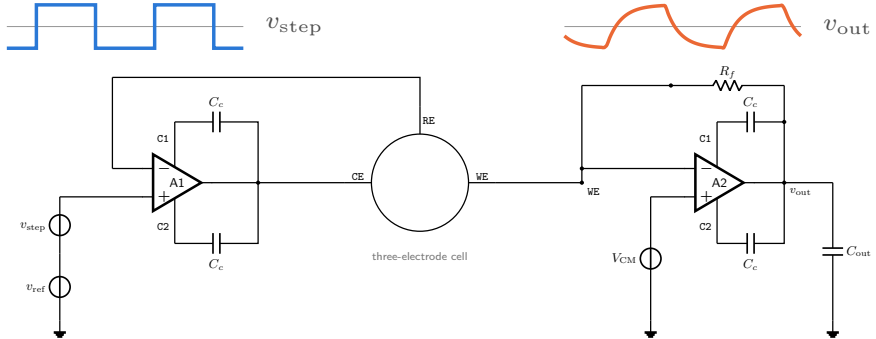
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Excite
Sense
Digitise
Solve

DAC: a small two-code square about the DC setpoint v_{ref}
 v_P across the cell; $i_{WE} = -v_{out}/R_f$
the SAR records both; firmware I/Q demodulation
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or a ratio to Z_{cal}

Magnitude and phase from codes: I/Q demodulation

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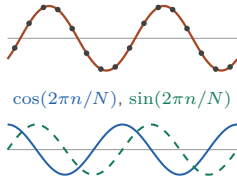
Vision

From currents to a score

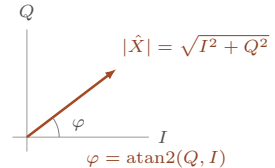
References

Backup

record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records v_P and v_{out}
Divide $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$
Ratio against Z_{cal} : R_f drops out

The four loads behind the EIS plots

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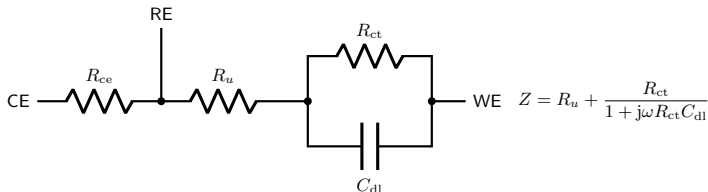
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Load	R_{ct}	C_{dl}	Gain code	Stands for
resistor	–	–	$N = 0$	an 11 k Ω resistor in place of the cell: no interface, so the trace is the channel's own error
fouled	10 k Ω	10 μ F	$N = 0$	a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface
typical	100 k Ω	1 μ F	$N = 20$	an electrode in use
fresh	1 M Ω	100 nF	$N = 63$	a new, clean surface

$R_u = R_{ce} = 1 \text{ k}\Omega$ throughout. Fouling: R_{ct} falls and C_{dl} rises through an electrode's life; tracking that drift is what EIS is for here.

Impedance spectroscopy on the same channel

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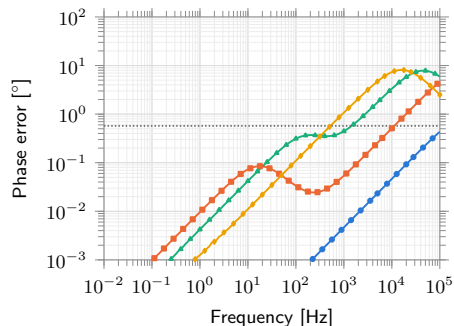
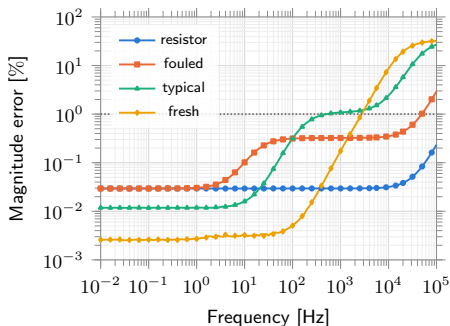
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Accuracy $\pm 1\%$, $\pm 0.6^\circ$ to 5 kHz; $\pm 5\%$ to 27 kHz. Limit: A1's loop bandwidth
 Range 246 Ω to 12.1 M Ω . Limits: TIA compliance below, one LSB of output above; 320 R_f codes in between

The excitation amplitude never enters: Z is the ratio of two records through one converter, so DAC gain and droop cancel.

What sets the EIS error, and what averaging buys

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$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/|Z|$ feedback resistor over the load: the gain code chosen

A_{eff} effective loop gain, ≈ 76 dB measured (0.66 % at $R_f/R = 40$); $A_{\text{OL}} \approx 83$ dB

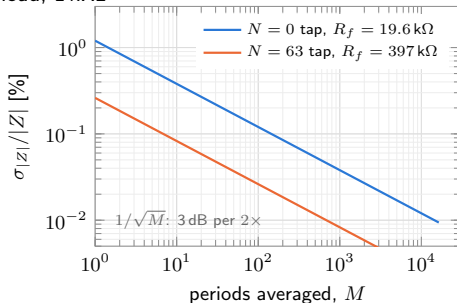
i_n front-end current noise per sample: 273 nA at $N=0$, 61 nA at $N=63$

V_1 excitation fundamental, $\frac{4}{\pi} \frac{A}{2}$

N, M samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the Z_{cal} ratio removes it. The random term is what remains, and it falls as $1/\sqrt{M}$.

Random term against periods averaged, 11 k Ω load, 1 kHz



Calibrating impedance

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$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency}$$

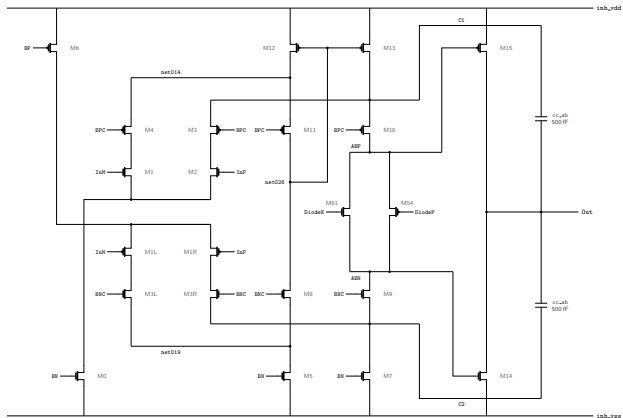
	On-chip poly, per decade	Off-chip standard, at test
Cancels	R_f , its pole, DAC, LSB, loop-gain floor	the same
Known to	$\sim 20\%$ chip to chip	$< 0.1\%$
Run	any time, in the field	once, stored in flash
Good for	trends: fouling is a change in one electrode's Z	absolute ohms

The poly need not be accurate, only stable. One stored number pins the scale; the on-chip ratio carries it through temperature and time.

The class-AB amplifier: one cell, both roles

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complementary
input pairs and
their cascodes

diode-connected
summing branchclass-AB
drive

output

testbench
compensation

The amplifier across corners, with mismatch σ

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Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	5.201

Bipolar accuracy through zero

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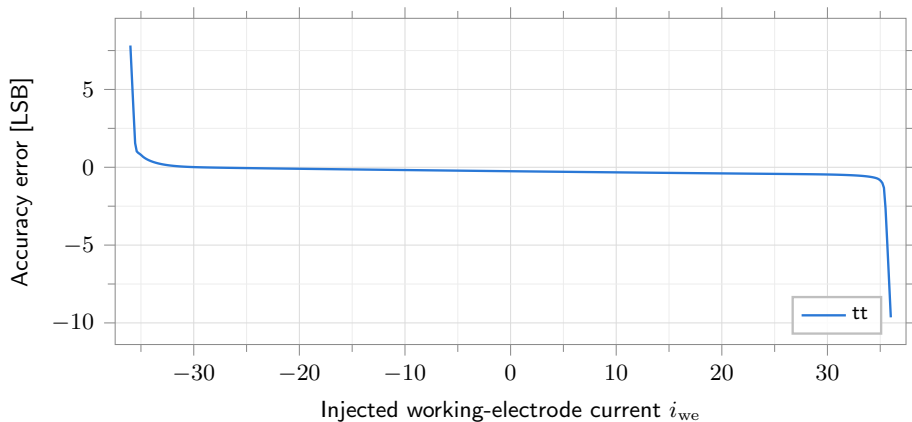
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Programmable feedback resistor: one string, 320 codes

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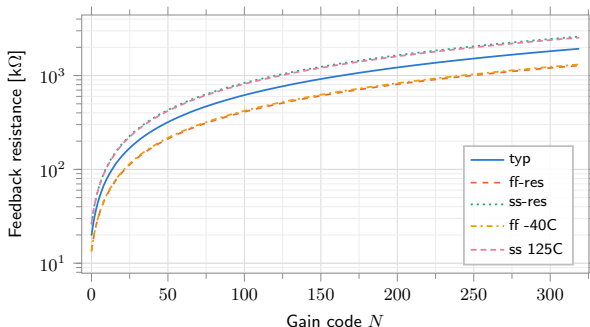
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- $R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$; full scale = $0.80 \text{ V}/R_f$, LSB = $1.59 \text{ mV}/R_f$
- **Max $\pm 40.7 \mu\text{A}$, min full scale $\pm 0.41 \mu\text{A}$, resolution 0.82 nA**
- Noise floor $i_n^2 = e_n^2 (1 + R_f/|Z_{el}|)^2/R_f^2 + 4kT/R_f$: 341 pA alone, 3.6–21 nA into an electrode (0.1–100 Hz, 560 k Ω)



N	R_f	Full scale	LSB
0	19.6 k Ω	$\pm 40.7 \mu\text{A}$	81 nA
63	397 k Ω	$\pm 2.02 \mu\text{A}$	4.0 nA
319	1.93 M Ω	$\pm 0.41 \mu\text{A}$	0.82 nA

Mismatch σ	σ/typ	
$R(N=0)$	71 Ω	0.36 %
$R(N=319)$	0.7 k Ω	0.04 %
Seam 63 $\rightarrow$ 64	504 Ω	6.8 %

10-bit SAR ADC, extracted netlist

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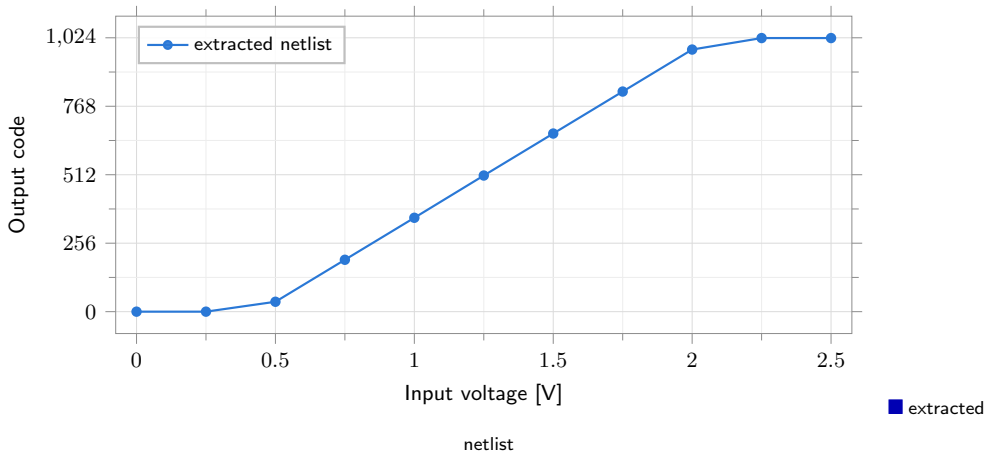
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23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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Parameter	Unit	tt	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

Reference DAC across corners, with mismatch σ

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Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000001
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	1.415

The front end, against the field

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	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Target application	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
The chip	4, simultaneous	1	1	1	1
Analog front end	bipolar	unipolar	bipolar	bipolar	bipolar
Operating modes	$I_{\max}$ (μA)	± 40.7	± 20	± 5	± 20
Analog blocks	Resolution (pA rms)	341 ^a	56.5	41	0.47
Digital	ADC step (pA)	820 ^b	153	n/a	n/a
Vision	Converter	10-bit SAR per channel	12-bit dual-slope	TDC	ADC
From currents to a score	Power (mW)	8.9 ^c	3.0	0.016	9.3
References	Process (nm)	65	65	180	350
Backup	On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none
	On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none

^aInput-referred noise, 0.1–100 Hz, ideal R_f , no electrode.

^bOne converter code at the maximum-gain tap; 81 nA at the lowest.

^cOne channel measuring: orchestrator + one tile (7.6 mW digital) + one analog channel (1.3 mW); the other tiles and channels gated, pads quiet.

Analog power: one channel

Castalia

Maxx Seminario

Target
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The chip

Analog front end

Operating modes

Analog blocks

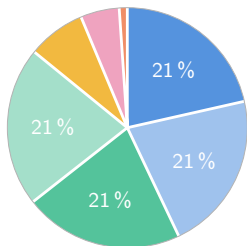
Digital

Vision

From currents to
a score

References

Backup



- A1, control amplifier, 244 μW , 21 %
- A2, transimpedance amplifier, 244 μW , 21 %
- ADC input buffer, 244 μW , 21 %
- debug buffer, 244 μW , 21 %
- two R-2R ladders, 89 μW , 8 %
- local bias generator, 59 μW , 5 %
- SAR converter, 12 μW , 1 %

Power

One channel	1.32 mW
Four channels	5.3 mW

Block diagram, again

Castalia

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Target application

The chip

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Analog blocks

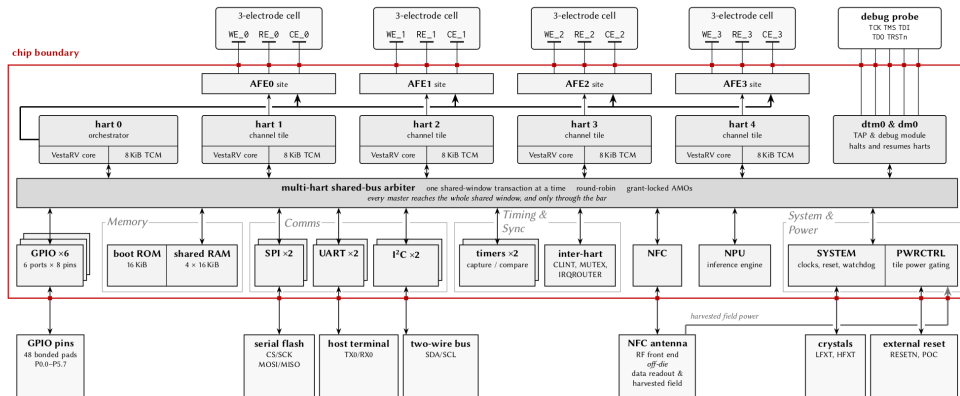
Digital

Vision

From currents to a score

References

Backup



Castalia at a glance

Castalia

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Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	12.8 mW core post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Digital, as closed

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Digital, post-route

Internal	8.69 mW
Switching	9.99 mW
Leakage	4.70 mW

Core logic	12.8 mW
Pad ring*	10.5 mW
Total	23.4 mW

One channel tile

Total	1.742 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726

Aggregate	1 314 802	1.260
------------------	------------------	--------------

SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2. *Pads at the statistical activity.

Power gating: one switched rail per channel tile

Castalia

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Analog blocks

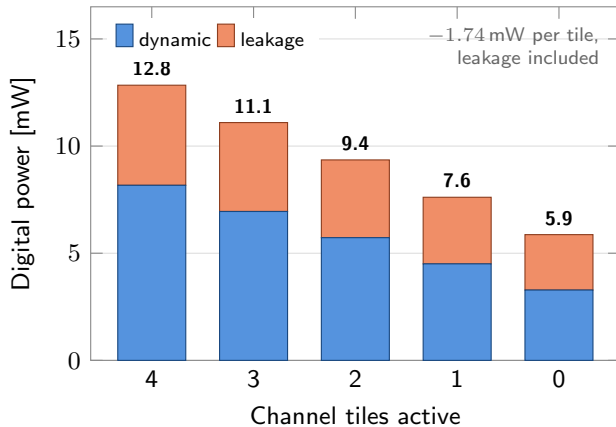
Digital

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Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.74 mW; the always-on band keeps 5.9 mW.

Low-power modes

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Mode	Digital	Analog	Total	What is off
Everything on, pads toggling	23.4 mW	5.3 mW	28.7 mW	nothing; pads at the statistical activity
Everything on, pads quiet	12.8 mW	5.3 mW	18.1 mW	pad switching; pad leakage is 0.03 mW
One channel measuring	7.6 mW	1.3 mW	8.9 mW	three tiles gated, with their channels
Orchestrator only	5.9 mW	0	5.9 mW	all four tiles and channels gated
Orchestrator asleep	2.6 mW	0	2.6 mW	its clock; always-on leakage remains
Deep sleep	< 2.6 mW	0	< 2.6 mW	shared RAM banks and NPU RAM powered off; NPU RAM alone leaks 0.66 mW

Digital: SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2.

Vision: the same channel, without the cartridge

Castalia

Maxx Seminario

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Analog blocks

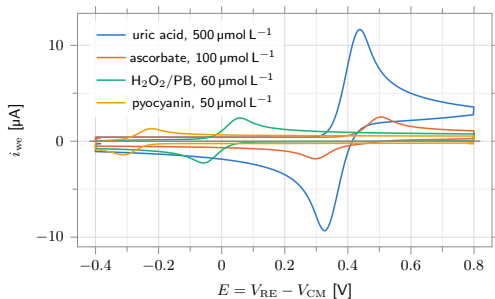
Digital

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Already on the die

4 sites	concurrent, each its own DAC, SAR, hart
EIS	electrode state without a second instrument
NFC	harvested boot, battery-free readout
MTCMOS	a parked site costs nothing
NPU	the decision stays on the patient

Left: a small-molecule wound panel, uric acid, ascorbate, H_2O_2 and pyocyanin, through the same channel.

The ovarian panel stays a serum test: CA-125 is an epitope on a 3–5 MDa mucin and does not reach interstitial fluid. The wearable is a different, reagentless panel on the same silicon; the implant is the same plus a fouling problem that EIS can watch but not solve.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential and baseline from each of the four sites
Mode	dense layer, one sigmoid; the logistic form of ROMA [6], four inputs
Weights	trained off-chip, staged by firmware
Output	one risk score

References (1/3)

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References

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References (3/3)

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Backup: electrode model validation

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References

Backup

Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF screening grades

Castalia

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Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.